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Systemic Screening of Strains of the Lion's Mane Medicinal Mushroom *Hericium erinaceus* (Higher Basidiomycetes) and Its Protective Effects on Aβ-Triggered Neurotoxicity in PC12 Cells

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ABSTRACT: *Hericium erinaceus* possesses multiple medicinal values. To date, however, there have been few studies of the systemic screening of *H. erinaceus* strains, and the neuroprotective effects of *H. erinaceus* prepared from homogenized, fresh fruiting bodies are not fully understood. In this study, 4 random primers were selected and used in random amplified polymorphic DNA (RAPD) polymerase chain reaction (PCR) to screen and evaluate the genetic diversity of 19 commercial strains of *H. erinaceus* from different localities in China. A total of 66 bands were obtained, and the percentage of polymorphic loci reached 80.30%. Five dendrograms were constructed based on RAPD by Jaccard cluster and within-group linkage analysis. Primer S20 as well as all 4 primers had great potential as specific primers for RAPD-PCR molecular identification and differentiation of *H. erinaceus* strains. Based on the results of submerged culture and fruiting body cultivation, strains HT-N, HT-J1, HT-C, and HT-M were identified as superior among the 19 *H. erinaceus* strains. Further study showed that the oral preparation of homogenized, fresh fruiting bodies of *H. erinaceus* could attenuate the Aβ25-35-triggered damage in PC12 cells by significantly increasing cell viability and by decreasing the release of lactate dehydrogenase. In conclusion, RAPD-PCR combined with liquid and solid cultures can be used well in the screening and identification of *H. erinaceus* strains, and products prepared from homogenized, fresh fruiting bodies of *H. erinaceus* had neuroprotective effects on PC12 cells.

KEY WORDS: medicinal mushrooms, *Hericium erinaceus*, lion's mane mushroom, neurodegeneration, RAPD-PCR, cluster analysis, Aβ, fresh fruiting bodies, liquid culture, solid cultivation, MTT, LDH

ABBREVIATIONS: AD, Alzheimer's disease; **Aβ,** β-amyloid; **DMSO,** dimethyl sulfoxide; **ITS,** internal transcribed spacer; **LDH,** lactate dehydrogenase; **MTT,** 3-(4, 5-dimethyl-2-thiazolyl)2,5-diphenyl-2H-tetrazolium bromide; **PCR,** polymerase chain reaction; **RAPD,** random amplified polymorphic DNA.

I. INTRODUCTION

Lion's mane mushroom, *Hericium erinaceus* (Bull.: Fr.) Pers. (Hydnaceae, higher Basidiomycetes), is a precious culinary-medicinal species and in East Asia has been used for hundreds of years as popular folk medicine to treat human gastrointestinal diseases. Reports have shown that both the fruiting bodies and mycelia of *H. erinaceus* contain various bioactive substances such as hericenones C-H,[1,2] erinacines A-I,[3–6] and polysaccharides. It possesses multiple biological functions such as antioxidation,[7] tumor inhibition,[8] hypolipidemic effects,[9] and antimicrobial[10] and antimetastasis activities.[11] Moreover, *H. erinaceus* can induce and enhance the synthesis and expression of nerve growth factor and thus may be used to treat and prevent dementia.[12,13] This is attractive because the incidence of human dementia has

increased sharply. Alzheimer disease (AD) is one of the most common dementias. Currently, there is no effective therapeutic method for treating AD, and new drugs or biological products for the efficient therapy or prevention of neurodegenerative diseases are in great demand.

Because of the broad medical value of *H. erinaceus*, finding an easy and reliable method for the identification of strains and further understanding its pharmacological properties on different diseases is greatly need. However, only a few studies have focused on the molecular phylogenetics and identification of *H. erinaceus*; the evolutionary history and relationship of *H. erinaceus* to other Basidiomycetes remains controversial. In addition, until now there has been no report on the neuroprotective effects of *H. erinaceus* prepared from homogenized fresh fruiting bodies. Adequate methods for the molecular identification and screening of *H. erinaceus* strains and the study of the effects of its fresh fruiting bodies on neuroprotection would be greatly appreciated. Compared with traditional methods of morphological and biochemical identification, new methods using molecular genetic markers, such as random amplified polymorphic DNA (RAPD) polymerase chain reaction (PCR), internal transcribed spacer (ITS), intersimple sequence repeat, amplified fragment-length polymorphism, restriction fragment length polymorphism, and sequence-related amplified polymorphism, are more sensitive and precise. RAPD-PCR, which is based on PCR with random primers, has been one of the commonly used molecular techniques to analyze the genetic relationships and diversities among different organisms. In addition to its easy and cost-effective properties, RAPD-PCR has 3 main advantages: (1) suitability for working with anonymous genomes; (2) applicability to problems for which only a limited amount of DNA is available; and (3) efficiency. By combining RAPD-PCR with liquid- and solid-culture screening, finding the right way to screen and identify superior *H. erinaceus* strains is possible. In addition, because β-amyloid (Aβ) peptide can trigger damage in neural cells and Aβ25–35 has been commonly used in the establishment of *in vitro* models of AD in neurotoxicity experiments,[14,15] Aβ-damaged PC12 cells can be a useful model for the study of the neuroprotective function of *H. erinaceus* fresh fruiting bodies.

The objectives of this study were (1) to characterize and find possible specific primers for the RAPD-PCR molecular identification and screening of 19 strains of *H. erinaceus* from China; (2) to screen and obtain superior strains of *H. erinaceus* by submerged culture and solid-substrate cultivation in addition to RAPD-PCR; and (3) to explore the protective effects of *H. erinaceus* prepared from homogenized fresh fruiting bodies on the attenuation of Aβ peptide neurotoxicity in PC12 cells.

II. MATERIALS AND METHODS

A. Strains, Cells, Media, and Reagents

H. erinaceus and *Cordyceps militaris* strains were purchased or gifted from different culture collections in China (Table 1). Mycelial stocks were maintained on a modified potato dextrose agar (PDA) slants containing 200 g potato/L, 20 g corn flour/L, 20 g glucose/L, 5 g peptone/L, 2 g KH_2PO_4/L, and 1.g 5 $MgSO_4$/L. Fungal inoculum was prepared from mycelia grown on PDA for 10 days at 24°C. Rat pheochromocytoma PC12 cells were a gift from the New Drug Research and Development Center of Chengdu Diao Groups (Chengdu, China). *H. erinaceus* oral preparation was obtained from Beijing Mushroom Science & Technology Co. Ltd. (Beijing, China). Fresh fruiting bodies were homogenized, spray dried, and prepared as a capsule. For treatment, the capsule was dissolved in dimethyl sulfoxide (DMSO) to prepare the stock solution with a final content of 10% (w/w). A lactate dehydrogenase (LDH) assay kit was purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, China). 3-(4,5-Dimethyl-2-thiazolyl)2,5-diphenyl-2H-tetrazolium bromide (MTT), DMSO, and Trypan blue were purchased from Amresco Co. Other chemicals were purchased from Chengdu Chemical Co. un-

TABLE 1: *Hericium erinaceus* Strains Used in This Study

Species	Strain*	Sources of Strains
H. erinaceus	HT-J1(3), HT-J2(4)	Beijing Mushroom Science & Technology Co. Ltd., Beijing, P.R. China
	HT-C(11)	Changbaishan Nature Reserve, Jilin Province, P.R. China
	HT-N(1)	Chinese Academy of Agricultural Sciences, Beijing, P.R. China
	HT-M(2)	Institute of Mianyang Edible Fungi, Sichuan Province, P.R. China
	HT-GH1(12), HT-GH2(13), HT-SH(17)	Institute of Jinxiang Edible Fungi, Shandong Province, P.R. China
	HT-12(16), HT-65(14), HT-66(15), HT-206(18), HT-208(19), HT-346(8), HT-496(9), HT-497(10)	Guangdong Culture Collection Center of Microbiology, Guangdong Province, P.R. China
	HT-Ha(5), HT-Hb(6), HT-Hc(7)	Institute of Sanming Fungi, Fujian Province, P.R. China
Cordyceps militaris	CM-m8	Institute of Jinxiang Edible Fungi, Shandong Province, P.R. China

*The number in parentheses is the coding number of *H. erinaceus* strains corresponding with Fig. 1A and B.

less otherwise indicated. All chemicals were of analytical grade.

B. RAPD-PCR and Cluster Analysis

Total genomic DNA was extracted using the cetyltrimethylammonium bromide method from ~3–5 g of mashed powder of dried *H. erinaceus* mycelia and quantified using a Beckman Du530 DNA/Protein Analyzer. For RAPD-PCR, 4 RAPD-PCR primers that had a high level of polymorphism and readability were selected from 22 random primers and used for RAPD amplification of *H. erinaceus* (Table 2). The RAPD-PCR reactions were carried out in a Bio-Rad GeneAmp PCR thermal cycler with the following parameters: 5 min of initial denaturing at 94°C, 45 cycles consisting of 1 min of denaturing at 94°C, 1 min of primer annealing at 36°C, 1 min of extension at 72°C, and a final 5-min extension at 72°C. PCR products were run on 1.5% agarose gel and then visualized and photographed using a Bio-imaging system from Gene Co.

The RAPD-PCR bands were scored to analyze the genetic relationships among strains and also build clustering analysis phenograms. A score of 1 was given if there was a band on the same migration position, 0 for no band, and 9 if the amplification failed or data were missed. The results were analyzed by Jaccard cluster and the within-group linkage method using SPSS software version 13.0.

C. Submerged Culture and Solid Cultivation of *H. erinaceus* Strains

For submerged culture, the mycelia of the 19 strains were grown in 50 mL PDA broth in a 250-mL shaking flask and incubated at 25°C at 160 rpm for 6 days. Later, the dry weight and morphological characteristics of the mycelia, including the shape, viability and dry weight of the pellets, smell, pH, transparence, and color of the broth were evaluated (the shape, viability, transparence, and color were qualified by visual observation).

The solid cultivation of *H. erinaceus* was performed using a general protocol[16] with few modifications. The basal substrate comprised 77% cotton seed hulls supplemented with 10% wheat bran, 10% rice bran, 2% gypsum powder, and

TABLE 2: Polymorphism of Random Amplified Polymorphic DNA Polymerase Chain Reaction Analysis Using Four Selected Primers

Primer	Sequence	Amplified Bands (n)	Polymorphic Loci (n)	Polymorphic Loci (%)
S20	GGACCCTTAC	17	12	70.59
S44	TCTGGTGAGG	15	13	86.67
S121	ACGGATCCTG	20	18	90.00
S252	TCACCAGCCA	14	10	71.43
Total		66	53	80.30

1% compound fertilizer (w/w). The water content was adjusted to 65% with a pH around 6.5. The substrates were packed in polypropylene bags with a wet weight of 200 ± 10 g and then autoclaved; when cooled, the spawn was inoculated and the inoculated bags were incubated at 25 ± 2°C in the dark, then flushed at 22 ± 2°C in ~90–95% humidity. The fruiting bodies were harvested, dried at 60°C to a constant weight, and weighed. Based on the quality of mycelia in submerged culture and fruiting production in solid cultivation, the preferred strains were obtained.

D. Aβ25-35 Induction and *H. erinaceus* Oral Preparation Treatment

PC12 cells were seeded in a 48-well plate at a density of 3×10^5 mL^{-1} in Dulbecco's modified Eagle's medium (Sigma) supplemented with 10% fetal bovine serum, penicillin (100 U · mL^{-1}), and streptomycin (100 μg · mL^{-1}). After incubation overnight to a confluence of about 80% at 37°C in a humidified atmosphere of 5% CO_2 (Thermo Forma), the cells were randomly divided into 3 groups: the normal group (no treatment), the model group (induced by Aβ25–35), and the test group (induced by Aβ25–35 after treatment with *H. erinaceus* oral preparation). Medium free of phenol red was added to each group; the medium for the test group contained an additional 1% concentration of *H. erinaceus* oral preparation. After 2 hours of incubation at 37°C, the cells in the model and test groups were exposed to Aβ25–35 to a final concentration of 10 μmol · L^{-1}. The effects of *H. erinaceus* oral preparation on Aβ25–35-induced neurotoxicity were performed by MTT assay, which is a rapid spectrophotometric assay for assessing cellular viability, whereby the yellow tetrazolium salt is reduced to insoluble purple formazan crystals by the mitochondria of viable cells.[12] After treatment with Aβ25–35 for 24 hours, 50 μL of the MTT stock solution (5 mg · mL^{-1}) diluted in serum and phenol red–free Dulbecco's modified Eagle's medium were added to PC12 cells, which then were incubated at 37°C in a 5% CO_2 atmosphere. Four hours later, the supernatants in each well were transferred separately into 1.5-mL Eppendorf tubes and stored at –20°C for LDH assay. Then, 500 μL DMSO was added to each well for 10 min to fully dissolve the formazan crystal. The absorbance of the reaction solution at 490 nm was measured by an Elx800 microplate reader (Biotek).

E. Effect of *H. erinaceus* on LDH Release by Treated PC12 Cells

An LDH assay kit was used to determine the release of LDH in the previously prepared supernatant of cell culture; the absorbance was measured at 440 nm using an Elx800 microplate reader (Biotek). The LDH content was calculated as:

$$\text{LDH content} = \frac{A_1 - A_2}{A_3 - A_4} \times \text{concentration of standard tube} \times 1000 \times \text{dilution},$$

where A_1, A_2, A_3, and A_4 are the absorbance of the sample, sample blank, standard, and the standard blank tubes, respectively.

F. Statistical Analysis

The experimental data were analyzed using a Student *t* test to evaluate significant differences. $P < 0.05$ was considered statistically significant.

III. RESULTS AND DISSCUSSION

A. RAPD-PCR and Cluster Analysis of 19 Strains of *H. erinaceus*

RAPD-PCR amplification of genomic DNA of the 19 strains of *H. erinaceus* using 4 selected primers yielded a total of 66 *H. erinaceus* DNA bands (Fig. 1A). The amplified fragments varied in size from 200 to 1900 bp, with an average of 16.5 polymorphic fragments per primer. A total of 53 polymorphic loci (80.30%), (Table 2), with the highest for primer S121.

The cluster results are shown in Fig. 1B. According to the Jaccard coefficient, the similarity coefficient among the 19 *H. erinaceus* strains ranged from 0.405 to 0.939, indicating the different degree of their genetic similarity. The clades of the 4 dendrograms produced by the different primers were quite divergent. Even though some strains, such as strains 4 and 3, were really clustered into one clade in all 4 dendrograms, most of the other strains were clustered into different clades. The dendrogram produced by primer S20 was closer to the dendrogram produced by all 4 primers than those of the other 3 dendrograms; strains 3 and 4; strains 5, 6, and 7; and strains 17, 18, and 19 were clustered, respectively, into the same clades of dendrograms I and V in Fig. 1C. *C. militaris* was, of course, clustered into a different clade (outgroup) and far away from *H. erinaceus* in all 4 dendrograms. From the dendrograms, we could obviously see that (1) strains from the same origin generally had higher genetic similarity; (2) some strains from different origins also had genetic similarity; and (3) several strains from the same localities had large genetic differences—this was the case for 9 strains from Guangdong Province (HT-12, HT-65, HT-66, HT-206, HT-207, HT-208, HT-346, HT-496, and HT-497), which were clustered into 3 categories, and 3 strains from Fujian Province (HT-Ha, HT-Hb, HT-Hc), which were clustered into 2 categories. These results may have the following possible explanations: (1) some strains may be variants or different names of the same strains; (2) by selling the same strain was distributed to different areas; (3) the strains of the same species were really different variants or subspecies. Also, by comparing dendrograms I and V in Fig. 1B, we concluded that both primer S20 and all 4 primers together can be specific primers for molecular identification of *H. erinaceus* strains using RAPD-PCR.

Based on ITS of *H. erinaceus*, previous reports showed that a pair of well-designed primers could be used in PCR identification of *H. erinaceus* species from Homobasidiomycetes as well as from different tissue types.[17] Based on our results, however, we can conclude that RAPD-PCR would offer more polymorphisms and be more useful in identifying and differentiating especially among different species or subspecies of *H. erinaceus*.

In addition to the RAPD-PCR technique, restriction fragment-length polymorphism, amplified fragment-length polymorphism, intersimple sequence repeat, ITS, and simple sequence repeat (microsatellite) also had been introduced in the investigation of inter- and intraspecies relationships, taxonomic classification, and evolutionary diversity among different species. These methods had been successfully used to discriminate *Pleurotus eryngii*[18] and *Lentinus edodes*[19] mushroom species as well as *Agaricus* spp.,[20] *Auricularia* spp.,[21] *Ganoderma* spp.,[22] and *Volvariella* spp.[23] Thanks to the extreme simplicity and requirement for no need of DNA sequence information and minimal amount of genomic DNA, RAPD-PCR was widely used in the distinction and analysis of the genetic diversity of mushroom species, including *Ganoderma lucidum* complex,[22] *Stropharia rugoso-annulata*,[24] and *Agaricus bisporus*.[25] Until now, however, there has been no report of RAPD-PCR used in the discrimination of *H. erinaceus*. Although RAPD-PCR has limitations both in stringent reaction conditions and poor reproducibility, the results of our study demonstrate the

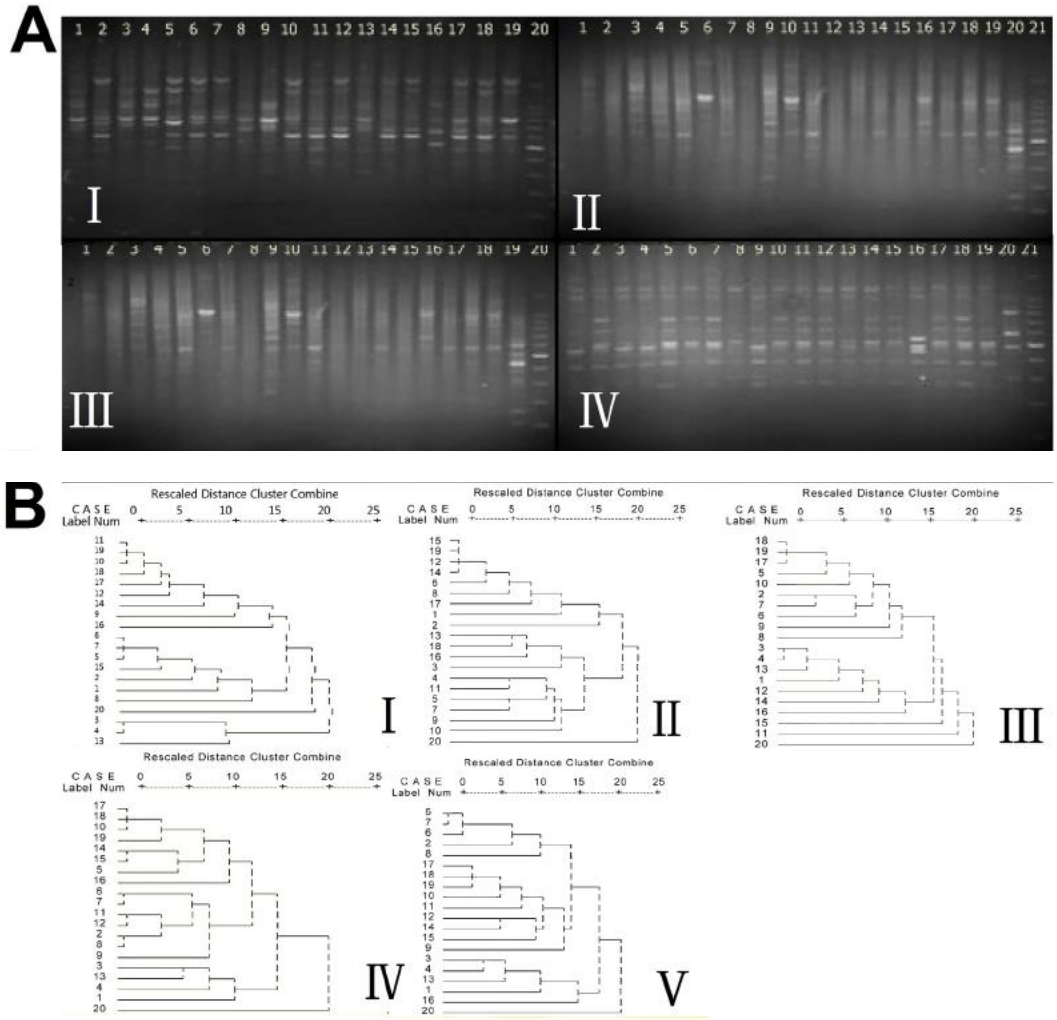

FIG. 1: A. Random amplified polymorphic DNA polymerase chain reaction (RAPD-PCR) amplification of *Hericium erinaceus* and *Cordyceps militaris* with 4 selected primers. I, II, III, and IV used primers S20, S44, S121, and S252, respectively. Lanes 1–19 (from left to right) are *H. erinaceus* strains HT-N, HT-M, HT-J1, HT-J2, HT-Ha, HT-Hb, HT-Hc, HT-346, HT-496, HT-497, HT-C, HT-GH1, HT-GH2, HT-65, HT-66, HT-12, HT-SH, HT-206, and HT-208, respectively. Lane 20 is a *C. militaris* strain as reference. M, 100-bp DNA molecular weight marker. **B.** Dendrograms of the cluster analysis of *H. erinaceus* and *C. militaris* RAPD-PCR with 4 selected primers. I, II, III, and IV used primers S20, S44, S121, and S252, respectively. V used primers S20, S44, S121, and S252 together. Cluster numbers 1–19 were *H. erinaceus* strains HT-N, HT-M, HT-J1, HT-J2, HT-Ha, HT-Hb, HT-Hc, HT-346, HT-496, HT-497, HT-C, HT-GH1, HT-GH2, HT-65, HT-66, HT-12, HT-SH, HT-206, and HT-208, respectively. Cluster number 20 is a *C. militaris* strain for reference.

great availability of RAPD-PCR markers in the discrimination of *H. erinaceus* strains. Nevertheless, we still need to emphasize that the combined use of several technologies would offer more precision and reliability for the identification of mushrooms.

B. *H. erinaceus* Strains Screening Using Submerged and Solid Substrate Cultivation

Submerged and solid-substrate cultivation are suitable ways to screen mushroom strains.[26] The

TABLE 3: Screening of *Hericium erinaceus* Strains by Liquid Culture

	Mycelial Pellets			Media of Liquid Culture			
Strain	**Shape**	**Viability***	**Average Dry Weight (g)**	**Color**	**Transparence***	**Smell**	**pH**
HT-N	Uniform circular balls	+++	1.47	Yellow-brown	+++	Pleasant	4.1
HT-M	Uniform circular balls	+++	1.34	Yellow-brown	++	Pleasant	4.1
HT-J1	Uniform circular balls	++	1.40	Yellow-brown	+++	Pleasant	4.1
HT-J2	Cotton-shaped	+++	1.46	Black	++	Pleasant	4.1
HT-Ha	Uniform circular balls	+	1.33	Yellow-brown	+++	—	5.1
HT-Hb	Nonuniform circular balls	++	1.15	Yellow-brown	+++	Pleasant	4.1
HT-Hc	Cotton-shaped	++	1.06	Yellow-brown	+	Pleasant	4.7
HT-346	Uniform circular balls	+++	1.26	Yellow-brown	++	Pleasant	4.1
HT-496	Nonuniform circular balls	+++	1.06	Yellow-brown	+	Pleasant	4.7
HT-497	Nonuniform circular balls	++	1.08	Yellow-brown	+++	Pleasant	4.1
HT-C	Cotton-shaped	+++	1.55	Yellow-brown	+++	—	5.1
HT-GH1	Uniform circular balls	++	1.14	Yellow-brown	+	Pleasant	4.7
HT-GH2	Nonuniform circular balls	+++	1.33	Black	+++	Pleasant	4.1
HT-65	Uniform circular balls	+	1.46	Yellow-brown	+++	—	5.1
HT-66	Nonuniform circular ball	+++	1.32	Yellow-brown	++	Pleasant	4.1
HT-12	Cotton-shaped	++	1.43	Yellow-brown	++	Pleasant	4.1
HT-SH	Cotton-shaped	+++	1.33	Yellow-brown	++	Pleasant	4.4
HT-206	Uniform circular balls	+++	1.23	Yellow-brown	+	Pleasant	4.7
HT-208	Uniform circular balls	++	1.19	Yellow-brown	++	—	5.1

*Presented as good (+), better (++), and best (+++) parameters of viability color and transparence based on visual observation. The results are averages of at least 2 replicates.

results of the submerged culture of *H. erinaceus* mycelial pellets are shown in Table 3. Most of the strains grew well in the liquid media, and mycelial pellets appeared as uniform circular balls with greater viability. In addition, the broths appeared to be a clear yellow-brown with a pleasant smell, and their pH ranged from 4.2 to 4.5. Based on the production of mycelial pellets and quality of fermentation, strains HT-C, HT-N, HT-J1, HT-J2, HT-M, HT-5, HT66, HT-346, HT-GH2, and HT-12 were ranked as superior *H. erinaceus* strains. The fruiting body production of *H. erinaceus* strains by solid cultivation is shown in Fig. 2. Most of the strains grew well; strains HT-N, HT-J1, HT-Ha, HT-C, HT-M, HT-GH1, and HT-Hc ranked ahead based on evaluation of the aver-

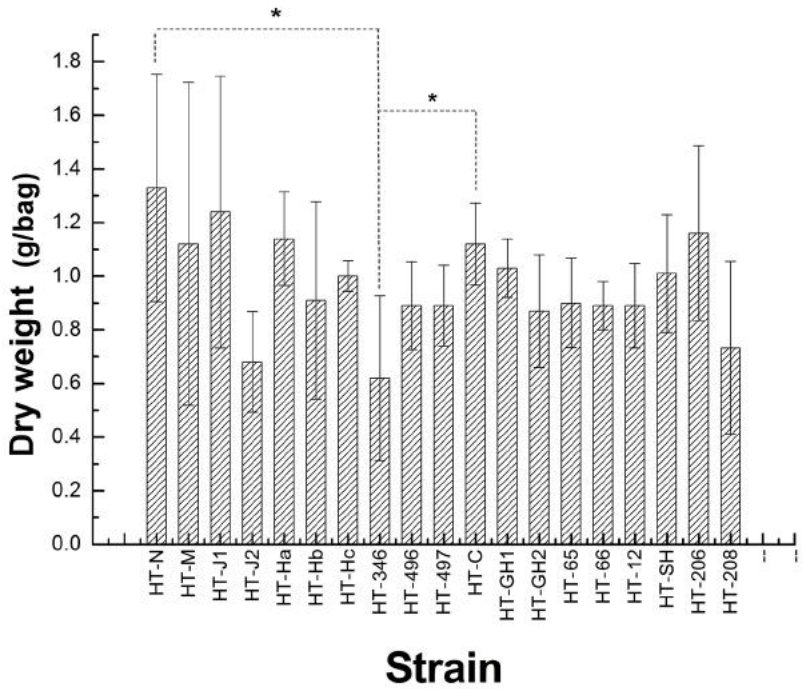

FIG. 2: Screening of *Hericium erinaceus* strains by solid cultivation. The graphs shows the dry weight of fruiting bodies (per bag) of 19 strains of *H. erinaceus*. At least 3 replicates per strain were tested. *$P < 0.05$.

age production. For example, the production of strains HT-N and HT-C was significantly higher than those of strain HT-346. Production of fruiting bodies in our study was not higher than the general productivity of *H. erinaceus*[27] as a result of the nonoptimal culture and fruiting conditions. Taking together the results of both liquid and solid cultivation, however, we still could conclude that strains HT-N, HT-J1, HT-C, and HT-M were superior *H. erinaceus* strains for future production and application. It should be noted that mycelial qualities and quantities of fruiting bodies are just 2 factors in the evaluation of *H. erinaceus*; for medicinal purposes, the contents of functional components are probably more important when screening strains.

C. Effects of *H. erinaceus* Oral Preparation on Aβ25–35-Triggered Damage in PC12 Cells

Because there are few reports of the neuroprotective effects of *H. erinaceus* products prepared from fresh fruiting bodies, here we tried to use MTT cell viability and LDH assay to test the effects of *H. erinaceus* products on Aβ-induced damage in PC12 cells. MTT is a general method for evaluating cell viability, and LDH is a reliable biochemical marker of damage to the cell plasma membrane.[28] In addition, because PC12 cells can extend processes similar to those produced by sympathetic neurons when exposed to nerve growth factor, Aβ-damaged PC12 cells can be the right model for this *in vitro* study.

As shown in Fig. 3, after treatment with Aβ25–35, the viability of PC12 cells in the model group significantly decreased ($P < 0.01$), whereas the viability of the test group pretreated with *H. erinaceus* oral preparation and later treated with Aβ25–35 was significantly increased ($P < 0.01$) compared with the normal control group. Furthermore, it was clear that cell viabilities in the test groups were significantly higher ($P < 0.01$) than those of the normal group, indicating that *H. erinaceus* could promote cell proliferation *in vi-*

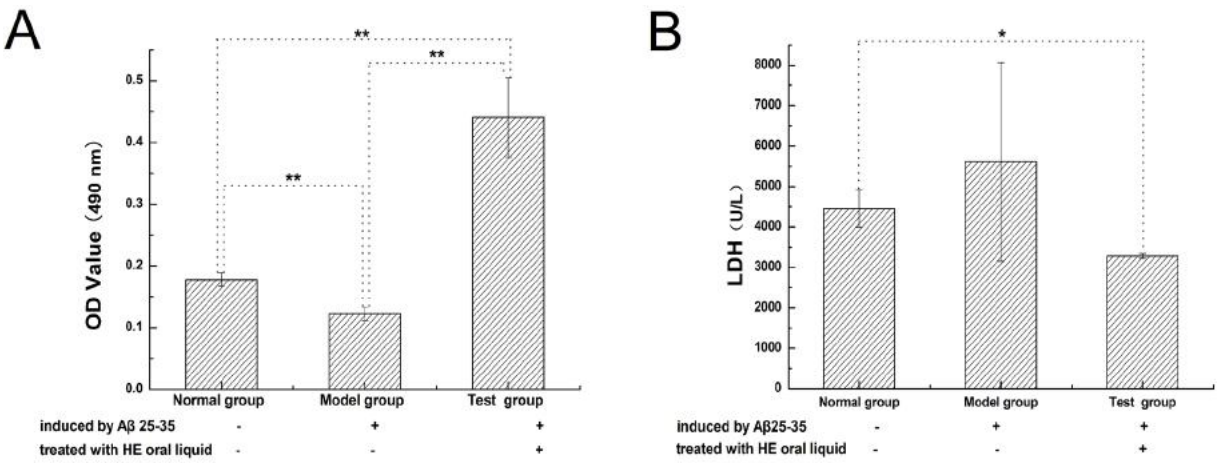

FIG. 3: Effects of *Hericium erinaceus* oral preparation on Aβ25–35-induced damage in PC-12 cells. **A.** Measurement of cell viabilities by MTT. **B.** Lactate dehydrogenase (LDH) assay of cell damage and protection. *$P < 0.05$; **$P < 0.01$. OD, optical density.

tro. These results were further validated by LDH assay of the cells. The exposure of PC12 cells to Aβ25–35 resulted in an increase in the release of LDH when compared with the normal group. Pretreatment with *H. erinaceus* at a 1% concentration for 2 hours significantly reduced Aβ-triggered LDH release ($P < 0.05$) compared with the normal control. Taken together, the findings of this study demonstrated that Aβ25–35-induced toxicity in PC12 cells could be significantly attenuated ($P < 0.05$) by pretreatment with an oral preparation of *H. erinaceus* homogenized fresh fruiting bodies. This conclusion is consistent with that of the study by Mori et al.,[29] who demonstrated that the lyophilized powder prepared from the fruiting bodies of *H. erinaceus* was both effective in improving mild cognitive impairment in human trials and preventing cognitive dysfunction in mice, indicating the great potential of *H. erinaceus* products in the prevention and therapy of neurodegeneration diseases.

IV. CONCLUSION

For the first time, RAPD–PCR was used to screen and evaluate the genetic diversity of *H. erinaceus*. Primer S20 as well as the 4 primers taken together were selected as the specific primers for RAPD-PCR molecular identification and differentiation of *H. erinaceus* strains. Based on liquid culture and solid cultivation, HT-N, HT-J1, HT-C, and HT-M were confirmed to be the superior *H. erinaceus* strains for future production and application. Oral preparation of *H. erinaceus* fresh fruiting bodies could attenuate Aβ25–35-triggered damage in PC12 cells. This is the first report regarding the neuroprotective functions of *H. erinaceus* prepared from homogenized fresh fruiting bodies. Further comprehensive studies of the identification, screening, and molecular neurological pharmacology of *H. erinaceus* are still needed.

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